

To demonstrate the reusability of the AI Circuit Breaker and Trust Engine across the digital lifecycle of an IoT-enabled electrocardiogram (ECG) device, decision logic must be strategically inserted at both **Design Time** (for semantic governance and model training) and **Run Time** (for real-time deterministic vetoes).

This "Observability-First" framework shifts the focus from simple model performance to a rigorous "Metrology of Trust," treating trustworthiness as a measurable property and ensuring that probabilistic AI interpretations of ECG rhythms remain strictly anchored in physical reality.

1. Design Time: Semantic Governance and Recursive Sanitization

At design time, decision logic is inserted into the **Semantic Knowledge Layer** and the **Recursive**

Learning Pipeline to build a governable, error-resistant foundation for the ECG device.

- **Establishing Semantic Guardrails (Ontologies):** Decision logic is hard-coded using W3C RDF/OWL ontologies to establish the "Immutable Truth" of the system. For an ECG device, these ontologies map out formalized "Impossible States" (e.g., physiologically impossible heart rates or conflicting conduction intervals) to constrain the AI's probabilistic inferences. This creates a "Knowledge Fabric" that standardizes clinical cardiology concepts into machine-readable deterministic rules.
- **Preventing Model Collapse (Data Exclusion):** In the training and continuous learning phases of the ECG lifecycle, decision logic acts as an "Immunology" layer. A Negative Learning/Exclusion Filter ensures that any telemetry or data associated with a previously "tripped" circuit breaker is flagged and purged from the training queue. This prevents the AI from ingesting synthetic errors (hallucinations) and reinforcing its own biases, effectively halting the feedback loop that leads to Model Collapse.
- **Scenario-Specific Vector Updates:** As the ECG device is tested or encounters new edge cases, logic reasoners evaluate whether proposed updates align with the ontological rules before modifying the domain-specific vector databases.

2. Run Time: Deterministic Veto and Real-Time Calibration

At run time, when the ECG device is deployed for patient monitoring (e.g., in a clinical setting or via a remote IoT wearable), decision logic is inserted dynamically at the point of execution to monitor the gap between the AI's diagnosis and the sensor's physical ground truth.

- **The Ground Truth Circuit Breaker:** The core run-time logic acts as a "Deterministic Veto," serving as the final gatekeeper that grants deterministic physical sensors (the ECG electrodes) absolute authority over the AI's probabilistic suggestions.
- **Trip Logic and Safe State Activation:** The operational workflow evaluates the discrepancy between the AI's intent and the physical reality using the following logic: IF (Intent Delta > Sensor Ground Truth Tolerance) OR (Trust Metric < Minimum Threshold) THEN ACTIVATE CIRCUIT BREAKER. If the device's AI outputs a "super-huberous" assertion (high confidence but lacking physical grounding, such as a false atrial fibrillation alert due to electrode motion artifacts), the circuit breaker trips. The system then defaults to a "Safe State," executing an emergency halt or reverting to manual human control (e.g., requiring a cardiologist's overread).

- **Real-Time Trust Metrics:** The decision logic continuously calculates quantifiable trust metrics to underwrite the system's reliability:
 - *Semantic Anomaly Score:* Quantifies the logical divergence between the AI's proposed interpretation and the established ECG domain ontology.
 - *Contextual Relevancy Index:* Evaluates if the AI's situational model is properly grounded in real-time environmental data (e.g., checking if sensors are accurately attached or if there is powerline interference).
- **Initialization & Semantic Handshake (SOP-01):** Before the ECG device begins active operation, run-time logic dictates that the agent must "ping" the Knowledge Graph to verify local constraints. If the sensor precision is not strictly higher than the operational tolerance, the device is prohibited from operating autonomously.

Reusability Across the Supply Chain and Device Lifecycle

By separating the architecture into distinct subsystems (Data Layer, Algorithmic Layer, LLM Layer, and Circuit Breaker Logic), this framework acts as a highly reusable agent.

Because the system manages the discrepancy between probabilistic AI and physical reality via a standardized **Model Context Protocol (MCP)**, the exact same Trust Engine can be deployed during the initial manufacturing and R&D of the ECG algorithm (design time), deployed to the edge on a portable IoT ECG device (run time), or utilized as a supervisory agent in a cloud-based clinical dashboard. Furthermore, the system includes a **Human-AI Calibration Coefficient** that monitors the human operator's cognitive load (via physiological state) to dynamically adjust circuit breaker thresholds, ensuring the human remains the ultimate dimension of trust across the entire lifecycle.